Notes: Hsu, Tian, and Xu (JFE, 2014)

Financial development and innovation: Cross-country evidence

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1 Big picture

- **Question.** How does financial development affect technological innovation, and do equity markets and credit markets affect innovation in the same way?
- **Setting.** The paper studies a panel of **32 developed and emerging economies** over **1976–2006**, with innovation measured using U.S. patent data and public-firm R&D data, and financial development measured separately for stock markets and banking systems.
- **Core challenge.** It is hard to identify the causal effect of finance on innovation because innovation and financial development can move together for reverse-causality reasons or because of omitted country characteristics.
- **Main idea.** Use a Rajan–Zingales style interaction design: ask whether industries that are *more dependent on external finance* or *more high-tech intensive* innovate relatively more in countries with deeper equity markets and relatively less in countries with deeper credit markets.
- **Main theoretical message.** Equity and credit are not interchangeable. Equity markets are better suited to finance risky, long-horizon, intangible, high-upside innovation. Credit markets can be less suitable because debt is collateral-based, banks are more conservative, and the feedback effects of market prices are weaker.
- **Main empirical result.** Industries that are more dependent on external finance and industries that are more high-tech intensive innovate more in countries with better developed equity markets, but innovate less in countries with better developed credit markets.
- **Important nuance.** If one pools equity and credit into a single "overall financial development" measure, the effects often weaken or disappear because the two components work in opposite directions.
- **Main patent-count result.** For industries with average external-finance dependence, moving from the 25th to the 75th percentile of equity market development raises relative patent counts by about **4.2%**, while moving from the 25th to the 75th percentile of credit market development lowers them by about **9.9%**.
- **Main high-tech result.** For industries with average high-tech intensiveness, the analogous comparison raises patent counts by about **3.0%** through equity markets and lowers them by about **6.6%** through credit markets.
- **Innovation quality result.** The same pattern shows up not only for patent counts but also for citations, originality, and generality.
- **Economic takeaway.** "More finance" is not the right summary. *What type of finance develops* matters for innovation.

2 Data and measurement (key objects)

2.1 Sample, countries, industries, and timing

- **Sample period.** 1976–2006.

- **Countries.** The sample covers 32 economies: Argentina, Australia, Austria, Belgium, Brazil, Canada, Denmark, Finland, France, Germany, Hungary, India, Ireland, Israel, Italy, Japan, Korea, Luxembourg, Malaysia, Mexico, Netherlands, New Zealand, Norway, Poland, Russia, Singapore, South Africa, Spain, Sweden, Switzerland, the UK, and the US.
- **Important sample choice.** The **US is excluded from the regression sample** to avoid a local-bias issue, because the paper uses U.S. patent filings to measure innovation and also uses U.S. data to build the industry mechanism variables.
- **Industries.** The baseline analysis focuses on manufacturing industries with two-digit SIC codes **20–39**, so there are **20 industries** in the main industry-level panel.
- **Why manufacturing?** Patents are most central and economically meaningful in manufacturing relative to many service industries.

2.2 Innovation measures

The paper constructs five innovation proxies.

- **Patent counts.** $\text{Patent}^n_{j,i,t}$ is the number of eventually granted patents in industry j, country i, and year t, based on the patent's *application year* rather than grant year.
- **Why application year matters.** Application year is closer to the timing of the actual invention than grant year.
- **Inventor-country assignment.** Patents are assigned to countries by *inventor nationality/location*, not assignee nationality, so the measure tracks where innovation activity occurs rather than where ownership ends up.
- **Government patents excluded.** The paper focuses on patents from individuals and non-government institutions, because government patenting may respond differently to financial development.
- **Citations.** $\text{Citation}^n_{j,i,t}$ counts forward citations to patents in a country-industry-year. This is the main quality measure.
- **Truncation correction.** Citation counts are adjusted using the Hall–Jaffe–Trajtenberg weighting approach, because later patents have had less time to accumulate citations.
- **Originality.** A patent's originality score is one minus the Herfindahl index of the technology-class distribution of the patents it cites. High originality means the invention draws on a broad base of prior technologies.
- **Generality.** A patent's generality score is one minus the Herfindahl index of the technology-class distribution of the later patents that cite it. High generality means the invention is useful for a broad set of later technologies.
- **R&D.** $\text{R\&D}^n_{j,i,t}$ is the sum of public firms' R&D expenses in a country-industry-year from Worldscope.

Important caution on R&D.

- It covers only *public* firms.

- Reporting quality varies across countries.
- Missing R&D does not necessarily mean zero innovation.
- For this reason the paper treats R&D as a supplementary proxy and interprets those estimates more cautiously than the patent-based estimates.

2.3 Scaling innovation by U.S. industry-year benchmarks

Following Rajan and Zingales (1998), the paper converts raw innovation measures into *relative* innovation measures by dividing each country-industry-year measure by the corresponding U.S. industry-year value. For example,

$$\text{Patent}_{j,i,t} = \frac{\text{Patent}^n_{j,i,t}}{\text{Patent}^n_{j,US,t}}. \tag{1}$$

Analogous scaling is used for citations, originality, generality, and R&D.

Interpretation.

- This removes time-varying industry-level patenting propensities common to all countries.
- It turns the innovation outcomes into relative measures: how innovative is country i's industry j relative to the same industry in the United States in the same year?

2.4 Financial development measures

The paper separates equity-market development from credit-market development.

$$\text{Equity}_{i,t} = \frac{\text{Stock Market Capitalization}_{i,t}}{\text{GDP}_{i,t}}, \tag{1}$$

$$\text{Credit}_{i,t} = \frac{\text{Bank Credit}_{i,t}}{\text{GDP}_{i,t}}. \tag{2}$$

It also considers an overall measure:

$$\text{Overall}_{i,t} = \frac{\text{Stock Market Capitalization}_{i,t} + \text{Bank Credit}_{i,t}}{\text{GDP}_{i,t}}. \tag{2}$$

Interpretation.

- **Equity** captures the depth of the stock market.
- **Credit** captures the depth of the banking sector.
- **Overall** is what many finance-growth papers would use, but in this paper it is partly a bad summary statistic because equity and credit often work in opposite directions.

Summary statistics that matter.

- In the pooled sample, mean **Equity** is **0.767** and mean **Credit** is **0.951**.

- Equity ranges widely across countries, from **0.136** in Poland to **1.774** in Switzerland (excluding the U.S.).
- Credit ranges from **0.282** in Russia to **2.548** in Japan.

2.5 Mechanism variables: external finance dependence and high-tech intensiveness

The identification strategy requires cross-industry characteristics that are plausibly technological or structural, not endogenous to each country's own financial development. The paper therefore measures the mechanism variables using **U.S. public-firm data**.

External finance dependence. For firm f in industry j, dependence on external finance is measured as

$$\mathrm{EFD}_{f,t} = \frac{\mathrm{CapEx}_{f,t} + \mathrm{R\&D}_{f,t} - \text{Cash Flow from Operations}_{f,t}}{\mathrm{CapEx}_{f,t} + \mathrm{R\&D}_{f,t}}. \tag{3}$$

Then:

- Compute the median across firms within a U.S. industry-year.
- Then take the time-series median over 1976–2006.
- This gives Dependence_j, a time-invariant industry characteristic.

High-tech intensiveness. The baseline proxy is based on firms' R&D growth. Specifically:

- For each U.S. public firm, compute the time-series median annual gross growth in R&D over 1976–2006.
- Then take the cross-sectional median within each industry.
- This yields High-tech_j.

Interpretation.

- **Dependence** captures how much an industry's investment model relies on outside funding.
- **High-tech** captures how R&D-intensive, risky, technologically frontier, and innovation-heavy an industry is.

Sample averages.

- Mean **Dependence**: **1.196**.
- Mean **High-tech**: **1.067**.

2.6 Other controls

The baseline regressions also include:

- **Value-Added**: industry j's share of manufacturing value added in country i, year t.

- **US-Export**: industry j's share of country i's exports to the U.S. in year t.

Why these controls?

- **Value-Added** absorbs within-country structural change across industries.
- **US-Export** matters because industries more exposed to the U.S. may have stronger incentives to patent in the U.S.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanisms and hypotheses

The paper develops two hypotheses.

Hypothesis 1: external finance dependence.

- Equity markets should help innovation more in industries that rely more on external finance.
- Credit markets should help less, and may even hurt, because debt financing is poorly suited to intangible, risky R&D projects and because banks provide weaker price signals than equity markets.

Hypothesis 2: high-tech intensiveness.

- Equity markets should especially help high-tech industries because they provide risk sharing, tolerate upside-skewed payoffs, and can support long-horizon risky projects.
- Credit markets should discourage innovation in high-tech industries because banks prefer collateralizable, safer, more conservative investments.

3.2 Baseline panel specification

The central empirical model is

$$\begin{aligned}\text{Innovation}_{j,i,t+1} = \beta_0 + \beta_1(\text{Equity}_{i,t} \times \text{Industry}_j) + \beta_2(\text{Credit}_{i,t} \times \text{Industry}_j) \\ + \beta_3\text{Value-Added}_{j,i,t} + \beta_4\text{US-Export}_{j,i,t} + \eta_{i,t+1} + \mu_j + \varepsilon_{j,i,t+1}.\end{aligned} \tag{3}$$

Here, Industry_j is either Dependence_j or High-tech_j.

Interpretation.

- $\eta_{i,t+1}$ are **country-year fixed effects**. These soak up all time-varying country-wide factors, including GDP level, aggregate development, institutions, reforms, and macro shocks.
- μ_j are **industry fixed effects**. These absorb time-invariant industry differences.
- Because country-year fixed effects are included, the coefficients are identified from *within-country, across-industry* differences in innovation.

- $\beta_1 > 0$ means equity market development helps innovation more in industries with higher external-finance dependence or higher high-tech intensity.
- $\beta_2 < 0$ means credit market development hurts innovation more in those same industries.

3.3 Overall-finance specification

The paper also estimates

$$\text{Innovation}_{j,i,t+1} = \delta_0 + \delta_1(\text{Overall}_{i,t} \times \text{Industry}_j) + \delta_2 \text{Value-Added}_{j,i,t} + \delta_3 \text{US-Export}_{j,i,t} + \eta_{i,t+1} + \mu_j + \varepsilon_{j,i,t+1}. \tag{4}$$

Interpretation.

- This is mainly a diagnostic equation.
- If equity and credit have offsetting effects, then δ_1 can be small or insignificant even when both β_1 and β_2 are economically important.

3.4 Why the interaction design is powerful

- The paper does *not* ask whether more financially developed countries innovate more on average.
- Instead it asks whether *the same country-year* exhibits relatively more innovation in industries that, according to U.S. benchmarks, should benefit more from a certain type of finance.
- This is a cleaner design because it differences out many country-wide confounds.

3.5 Alternative empirical specifications

The paper checks robustness using:

- country-industry fixed effects,
- standard errors clustered only by country,
- alternative proxies for equity development (stock market traded value/GDP) and credit development (private credit/GDP),
- an alternative high-tech proxy based on market valuation of R&D,
- innovation measures defined at the three-digit technology-class level rather than the two-digit SIC level.

3.6 Interpretation of coefficients

The paper repeatedly converts interaction coefficients into economically meaningful effects by multiplying:

- the coefficient estimate,
- the sample mean of the industry mechanism variable, and

- the interquartile difference in the financial development measure.

For example, the patent-count effect under external finance dependence is computed as

$$0.047 \times 1.196 \times 0.749 \approx 0.042, \tag{4}$$

which the paper reads as a **4.2%** increase in relative patenting when moving from the 25th to the 75th percentile of equity-market development.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the estimates

The paper does *not* use a traditional external instrument. Its identification comes from a structured interaction design.

- Country-year fixed effects remove any country-wide shock in a given year.
- Industry fixed effects remove permanent industry differences.
- U.S.-based industry characteristics are treated as technological benchmarks that are predetermined with respect to non-U.S. country-year shocks.
- The coefficients are therefore identified from whether financially developed countries have *differential* innovation in industries that should be more externally finance dependent or more high-tech intensive.

4.2 Why use U.S. industry characteristics?

- Following Rajan and Zingales (1998), the U.S. is treated as a relatively frictionless and informative benchmark.
- This helps isolate industry *technology* from country-specific financing conditions.
- Making the mechanism variables time invariant is also important: it prevents the interaction terms from mechanically reflecting contemporaneous country-specific shocks to industries.

4.3 Why country-year fixed effects matter so much

Without country-year fixed effects, richer countries with better institutions, better education, better legal environments, and stronger innovative cultures could mechanically appear to validate a finance-innovation link.

With country-year fixed effects:

- the comparison is inside a country-year,
- only cross-industry differential effects matter,
- the design is much closer to asking whether finance helps the industries that theory says should need it most.

4.4 Why separate equity from credit?

This is one of the paper's sharpest points.

- If one bundles stock-market depth and bank-credit depth into one aggregate finance measure, the two offsetting channels can obscure the true effects.
- The paper shows exactly this: overall financial development is often insignificant because equity and credit move innovation in opposite directions.

4.5 What makes the exercise credible

- The signs line up tightly with the paper's mechanism stories.
- The results show up across patent quantity, patent quality, and patent breadth measures.
- They survive country-industry fixed effects, alternative clustering, alternative finance measures, alternative high-tech measures, and technology-class definitions.
- The fact that the overall-finance measure often washes out is actually a strength, not a weakness: it is exactly what one should expect if equity and credit work differently.

4.6 Main limitations

- The analysis is at the country-industry-year level, not the firm level, so it does not pin down which firms within an industry are driving the effects.
- Patenting is measured using U.S. patents, which is standard but still imperfect for global innovation.
- R&D data are public-firm only and noisy outside the U.S.
- The U.S.-benchmark assumption may be imperfect if technology rankings differ systematically across countries.
- The design is persuasive about differential effects, but it does not fully separate all possible channels through which equity and credit markets differ.

4.7 Relation to the literature

- Relative to the classic finance-growth literature, the paper shifts attention from growth to *innovation* as the real outcome.
- Relative to Rajan and Zingales (1998), it keeps the interaction logic but applies it to innovation rather than value-added growth.
- Relative to later finance-and-innovation work, its main contribution is the strong distinction between *equity-market development* and *credit-market development*.

5 Tables

5.1 Table 1: summary statistics

Table 1
Summary statistics.

Panel A reports the country-level averages, including the pooled averages of innovation variables (*Patent**, *Citation**, *Originality**, *Generality**, and *R&D**), *Value-Added*, and *US-Export*, and the time series averages of *Equity* and *Credit*. *Patent** is defined as the number (in thousands) of patents in industry *j* filed by non-government institutions or individuals of country *i* to the United State Patent and Trademark Office (USPTO) in year *t*. *Citation** is defined as the number (in thousands) of patents that cite the patents in industry *j* filed by non-government institutions or individuals of country *i* to the USPTO in year *t*. *Originality** (*Generality**) is defined as the sum of the originality (generality) scores (in thousands) of patents in industry *j* filed by non-government institutions or individuals of country *i* to the USPTO in year *t*. *R&D** is defined as the sum of all research and development (R&D) expenses (in millions of US dollars) reported by public firms in industry *j* of country *i* in year *t*. *Equity* (*Credit*) is the ratio of stock market capitalization (domestic bank credit) over gross domestic product (GDP). *Value-Added* denotes the share of industry *j*'s value-added in country *i* in year *t*. *US-Export* denotes the share of industry *j*'s in country *i*'s export to the US in year *t*. Industry is defined by two-digit standard industry classification (SIC) codes. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976–2006.

Panel A Country	*Patent**	*Citation**	*Originality**	*Generality**	*R&D**	*Equity*	*Credit*	*Value-Added*	*US-Export*
Argentina	0.002	0.021	0.000	0.000	0.000	0.273	0.369	0.050	0.047
Australia	0.033	0.297	0.004	0.003	0.008	0.821	0.705	0.047	0.044
Austria	0.029	0.189	0.002	0.002	0.007	0.183	1.102	0.050	0.049
Belgium	0.036	0.295	0.006	0.004	0.042	0.557	0.916	0.075	0.049
Brazil	0.004	0.024	0.001	0.000	0.014	0.284	0.792	0.075	0.046
Canada	0.144	1.620	0.023	0.017	0.095	0.821	1.188	0.047	0.047
Denmark	0.020	0.158	0.002	0.002	0.029	0.470	0.796	0.047	0.045
Finland	0.029	0.269	0.005	0.003	0.129	0.853	0.627	0.048	0.047
France	0.220	1.797	0.025	0.027	0.640	0.579	1.018	0.048	0.047
Germany	0.731	5.471	0.076	0.081	0.873	0.370	1.123	0.047	0.046
Hungary	0.019	0.084	0.001	0.001	0.388	0.802	0.048	0.048	0.049
India	0.013	0.053	0.002	0.000	0.007	0.341	0.477	0.045	0.043
Ireland	0.006	0.071	0.002	0.001	0.007	0.617	0.736	0.046	0.047
Israel	0.035	0.422	0.008	0.005	0.008	0.483	1.087	0.054	0.048
Italy	0.096	0.635	0.010	0.009	0.286	0.325	0.936	0.047	0.048
Japan	2.132	23.002	0.337	0.347	1.805	0.828	2.548	0.046	0.046
Korea	0.115	0.853	0.022	0.009	0.095	0.470	0.574	0.046	0.048
Luxembourg	0.003	0.016	0.000	0.000	0.000	1.550	0.946	0.071	0.051
Malaysia	0.002	0.017	0.001	0.000	0.002	1.624	1.125	0.045	0.046
Mexico	0.004	0.026	0.001	0.000	0.000	0.263	0.436	0.047	0.045
Netherlands	0.072	0.641	0.011	0.010	0.190	0.921	1.267	0.048	0.044
New Zealand	0.005	0.037	0.001	0.000	0.000	0.406	0.734	0.123	0.046
Norway	0.010	0.076	0.001	0.001	0.012	0.359	0.638	0.047	0.047
Poland	0.003	0.019	0.000	0.000	0.000	0.136	0.375	0.052	0.049
Russia	0.039	0.248	0.002	0.003	0.001	0.277	0.282	0.047	0.051
Singapore	0.011	0.131	0.004	0.002	0.004	1.611	0.635	0.047	0.047
South Africa	0.006	0.051	0.001	0.001	0.003	1.578	1.211	0.055	0.047
Spain	0.013	0.073	0.001	0.001	0.001	0.541	1.084	0.046	0.047
Sweden	0.063	0.591	0.007	0.007	0.328	0.847	0.984	0.047	0.046
Switzerland	0.092	0.741	0.008	0.011	0.390	1.774	1.632	0.074	0.047
UK	0.203	1.967	0.029	0.029	0.426	1.283	0.972	0.046	0.046
US	4.167	60.604	0.958	0.745	3.579	1.093	1.635	0.046	–

Panel B reports industry-level averages, including the pooled averages of innovation variables (*Patent**, *Citation**, *Originality**, *Generality**, and *R&D**), time series averages of the dependence on external finance (*Dependence*) and high-tech intensiveness (*High-tech*), and pooled averages of control variables (*Value-Added* and *US-Export*).

Panel B SIC 2	Industry	*Patent**	*Citation**	*Originality**	*Generality**	*R&D**	*Dependence*	*High-tech*	*Value-Added*	*US-Export*
20	Food and Kindred Products	0.063	0.584	0.007	0.006	0.232	1.174	1.064	0.074	0.047
21	Tobacco Products	0.006	0.059	0.001	0.000	0.137	1.290	1.022	0.005	0.000
22	Textile Mill Products	0.012	0.103	0.001	0.001	0.030	1.131	1.048	0.017	0.015
23	Apparel and Other Finished Products Made from Fabrics and Similar Materials	0.006	0.053	0.000	0.000	0.014	1.474	1.188	0.052	0.057
24	Lumber and Wood Products, Except Furniture	0.010	0.102	0.001	0.001	0.016	1.177	1.109	0.028	0.011
25	Furniture and Fixtures	0.035	0.401	0.004	0.003	0.079	1.241	1.088	0.033	0.025
26	Paper and Allied Products	0.173	2.084	0.033	0.027	0.077	1.104	1.018	0.039	0.021
27	Printing, Publishing, and Allied Industries	0.020	0.232	0.004	0.003	0.041	1.124	1.011	0.026	0.003
28	Chemicals and Allied Products	1.164	10.796	0.180	0.144	2.391	1.028	1.120	0.057	0.084
29	Petroleum Refining and Related Industries	0.156	1.481	0.029	0.023	0.390	1.035	0.975	0.010	0.002
30	Rubber and Miscellaneous Plastics Products	0.080	0.804	0.013	0.011	0.193	1.203	1.029	0.047	0.035
31	Leather and Leather Products	0.002	0.028	0.000	0.000	0.025	1.237	1.075	0.010	0.019
32	Stone, Clay, Glass, and Concrete Products	0.087	0.897	0.016	0.013	0.074	1.102	1.062	0.025	0.013
33	Primary Metal Industries	0.073	0.669	0.011	0.009	0.206	1.203	1.014	0.064	0.076
34	Fabricated Metal Products, Except Machinery and Transportation Equipment	0.110	1.124	0.015	0.013	0.110	1.286	1.051	0.118	0.099

Table 1 (*continued*)

Panel B reports industry-level averages, including the pooled averages of innovation variables (*Patent**, *Citation**, *Originality**, *Generality**, and *R&D**), time series averages of the dependence on external finance (*Dependence*) and high-tech intensiveness (*High-tech*), and pooled averages of control variables (*Value-Added* and *US-Export*).

Panel B SIC 2	Industry	*Patent**	*Citation**	*Originality**	*Generality**	*R&D**	*Dependence*	*High-tech*	*Value-Added*	*US-Export*
35	Industrial and Commercial Machinery and Computer Equipment	1.166	17.017	0.285	0.220	1.390	1.126	1.071	0.118	0.131
36	Electronic and Other Electrical Equipment and Components, except Computer Equipment	1.228	17.956	0.303	0.240	2.502	1.168	1.110	0.090	0.097
37	Transportation Equipment	0.675	7.706	0.116	0.098	2.749	1.309	1.102	0.049	0.043
38	Measuring, Analyzing, and Controlling Instruments; Photographic, Medical and Optical Goods; Watches and Clocks	0.681	9.797	0.138	0.111	1.082	1.205	1.098	0.050	0.068
39	Miscellaneous Manufacturing Industries	0.057	0.735	0.005	0.005	0.086	1.304	1.085	0.065	0.072

Panel C reports the summary statistics of variables across all country–industry–year observations. *Patent*, *Citation*, *Originality*, *Generality*, and *R&D* are country–industry–year numbers of patents, citation, originality, generality, and R&D, respectively, scaled by the corresponding industry–year values in US data.

Panel C Variable	Mean	Standard deviation	25%	Median	75%
Patent	0.057	0.184	0.001	0.004	0.021
Citation	0.050	0.178	0.001	0.003	0.015
Originality	0.049	0.179	0.000	0.002	0.012
Generality	0.054	0.186	0.000	0.002	0.014
R&D	0.205	0.536	0.001	0.016	0.133
Equity	0.767	0.711	0.299	0.546	1.048
Credit	0.951	0.511	0.550	0.890	1.195
Dependence	1.196	0.102	1.125	1.189	1.263
High-tech	1.067	0.047	1.026	1.068	1.100
Value-Added	0.050	0.056	0.017	0.035	0.064
US-Export	0.047	0.063	0.007	0.023	0.064

Read: Table 1 tells you what the sample looks like at the country level, industry level, and pooled country-industry-year level.

- The pooled averages of the main innovation proxies are:
  - Patent: **0.057**
  - Citation: **0.050**
  - Originality: **0.049**
  - Generality: **0.054**
  - R&D: **0.205**
- The pooled means of the finance variables are:
  - Equity: **0.767**
  - Credit: **0.951**
- The pooled means of the mechanism variables are:
  - Dependence: **1.196**
  - High-tech: **1.067**
- The most innovative industries in patent counts are SIC **36**, **35**, and **28**.

- The industries with the highest R&D are SIC **37**, **36**, and **28**.

Why it matters. The table shows there is large cross-country and cross-industry variation to identify the interaction effects. It also makes clear that the benchmark country for scaling, the U.S., is an outlier in patenting.

5.2 Table 2: dependence on external finance and innovation

Read: Table 2 is the core test of Hypothesis 1.

- **Patent counts, row (3):** $\beta_1 = 0.047$ and $\beta_2 = -0.128$. Interpretation: equity markets raise innovation in externally finance-dependent industries, while credit markets reduce it.
- **Economic magnitude:** for an average-dependence industry, patent counts rise by about **4.2%** when equity development moves from the 25th to the 75th percentile, but fall by about **9.9%** when credit development moves from the 25th to the 75th percentile.
- **Citations, row (7):** $\beta_1 = 0.033$, $\beta_2 = -0.087$. That is, the quality of innovation moves in the same direction as the quantity.
- **Originality, row (11):** $\beta_1 = 0.026$, $\beta_2 = -0.059$.
- **Generality, row (15):** $\beta_1 = 0.025$, while the credit coefficient is negative but not statistically significant.
- **R&D, rows (17)–(20):** largely insignificant. This is one reason the paper treats R&D as a weaker measure in this setting.
- **Overall finance, rows (4), (8), (12), (16), (20):** mostly insignificant. This is exactly the offsetting-forces point.

Why it matters. Table 2 is the cleanest evidence that equity and bank-credit development are not just different in degree, but different in kind.

Table 2
Financial development, dependence on external finance, and innovation.
This table report the results estimating various forms of $Innovation_{ijJ,t+1} = \beta_0 + \beta_1\,(Equity_{iJ} \times Dependence_j) + \beta_2\,(Credit_{iJ} \times Dependence_j) + \beta_3\,Value\text{-}Added_{ijJ,t} + \beta_4\,US\text{-}Export_{ijJ,t} + \eta_{i,t+1} + \mu_j + \varepsilon_{ijJ,t+1}$ and $Innovation_{ijJ,t+1} = \delta_0 + \delta_1(Overall_{iJ} \times Dependence_j) + \delta_2 Value\text{-}Added_{ijJ,t} + \delta_3 US\text{-}Export_{ijJ,t} + \eta_{i,t+1} + \mu_j + \varepsilon_{ijJ,t+1}$. $Innovation_{ijJ,t+1}$ is the relative innovation measure in industry j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{iJ}$ denotes country i's stock market capitalization over GDP in year t, and $Credit_{iJ}$ denotes country i's bank credit over GDP in year t. $Overall_{iJ}$ denotes country i's stock market capitalization plus bank credit over GDP in year t. $Dependence_j$ denotes two-digit SIC industry j's dependence on external financing. $Value\text{-}Added_{ijJ}$ denotes the share of industry j's value-added in country i in year t. $US\text{-}Export_{ijJ}$ denotes the share of industry j's in country i's export to the US in year t. $\eta_{i,t+1}$ denotes the dummies for country i in year $t+1$, and μ_j denotes the dummies for industry j. $\varepsilon_{ijJ,t+1}$ denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country and industry. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976 – 2006.

Innovation	$Equity_{iJ} \times Dependence_j$	$Credit_{iJ} \times Dependence_j$	$Overall_{iJ} \times Dependence_j$	*Value-Added*	*US-Export*	*Adj-R^2*	*Obs.*
Patent (1)	0.013**			0.005	0.061**	0.829	7548
	(0.006)			(0.013)	(0.025)		
Patent (2)		−0.115**		0.006	0.053**	0.835	7434
		(0.058)		(0.013)	(0.024)		
Patent (3)	0.047***	−0.128**		0.006	0.055**	0.836	7354
	(0.018)	(0.062)		(0.013)	(0.024)		
Patent (4)			−0.037	0.004	0.058**	0.831	7354
			(0.024)	(0.013)	(0.025)		
Citation (5)	0.012**			0.009	0.032***	0.763	7548
	(0.005)			(0.009)	(0.012)		
Citation (6)		−0.077**		0.010	0.026**	0.767	7434
		(0.033)		(0.009)	(0.012)		
Citation (7)	0.033***	−0.087**		0.011	0.028**	0.767	7354
	(0.010)	(0.035)		(0.010)	(0.012)		
Citation (8)			−0.023	0.009	0.030**	0.764	7354
			(0.015)	(0.009)	(0.012)		
Originality (9)	0.011**			0.011	0.035***	0.773	7548
	(0.004)			(0.008)	(0.013)		
Originality (10)		−0.051*		0.012	0.031**	0.775	7434
		(0.026)		(0.008)	(0.013)		
Originality (11)	0.026***	−0.059**		0.012	0.033**	0.775	7354
	(0.009)	(0.028)		(0.009)	(0.013)		
Originality (12)			−0.014	0.011	0.034***	0.773	7354
			(0.012)	(0.008)	(0.013)		
Generality (13)	0.008*			0.010	0.041***	0.801	6814
	(0.004)			(0.009)	(0.015)		
Generality (14)		−0.066		0.010	0.037**	0.804	6700
		(0.042)		(0.009)	(0.014)		
Generality (15)	0.025**	−0.073		0.010	0.039***	0.804	6620
	(0.011)	(0.044)		(0.010)	(0.015)		
Generality (16)			−0.021	0.009	0.041***	0.802	6620
			(0.017)	(0.009)	(0.015)		
R&D (17)	0.714			42.019	28.678	0.236	3582
	(0.500)			(45.099)	(34.888)		
R&D (18)		−0.877		48.965	15.954	0.238	3561
		(1.235)		(43.328)	(36.702)		
R&D (19)	0.943	−1.008		45.516	20.709	0.239	3536
	(0.606)	(1.286)		(44.846)	(37.820)		
R&D (20)			−0.216	45.361	20.340	0.236	3536
			(0.638)	(43.899)	(36.719)		

Table 3
Financial development, high-tech intensiveness, and innovation.
This table reports the results estimating various forms of $Innovation_{ijJ,t+1} = \beta_0 + \beta_1(Equity_{iJ} \times High\text{-}tech_j) + \beta_2(Credit_{iJ} \times High\text{-}tech_j) + \beta_3\,Value\text{-}Added_{ijJ,t} + \beta_4\,US\text{-}Export_{ijJ,t} + \eta_{i,t+1} + \mu_j + \varepsilon_{ijJ,t+1}$ and $Innovation_{ijJ,t+1} = \delta_0 + \delta_1(Overall_{iJ} \times High\text{-}tech_j) + \delta_2 Value\text{-}Added_{ijJ,t} + \delta_3 US\text{-}Export_{ijJ,t} + \eta_{i,t+1} + \mu_j + \varepsilon_{ijJ,t+1}$. $Innovation_{ijJ,t+1}$ is the relative innovation measure in industry j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{iJ}$ denotes country i's stock market capitalization over GDP in year t, and $Credit_{iJ}$ denotes country i's bank credit over GDP in year t. $Overall_{iJ}$ denotes country i's stock market capitalization plus bank credit over GDP in year t. $High\text{-}tech_j$ denotes two-digit SIC industry j's median R&D growth in the US. $Value\text{-}Added_{ijJ}$ denotes the share of industry j's value-added in country i in year t. $US\text{-}Export_{ijJ}$ denotes the share of industry j's in country i's export to the US in year t. $\eta_{i,t+1}$ denotes the dummies for country i in year $t+1$, and μ_j denotes the dummies for industry j. $\varepsilon_{ijJ,t+1}$ denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country and industry. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976–2006.

Innovation	$Equity_{iJ} \times High\text{-}tech_j$	$Credit_{iJ} \times High\text{-}tech_j$	$Overall_{iJ} \times High\text{-}tech_j$	*Value-Added*	*US-Export*	*Adj-R^2*	*Obs.*
Patent (1)	0.014***			0.003	0.022***	0.918	7548
	(0.005)			(0.003)	(0.004)		
Patent (2)		−0.085***		0.002	0.025***	0.920	7434
		(0.016)		(0.003)	(0.004)		
Patent (3)	0.038***	−0.096***		0.003	0.025***	0.920	7354
	(0.007)	(0.016)		(0.003)	(0.004)		
Patent (4)			−0.026***	0.002	0.025***	0.919	7354
			(0.008)	(0.003)	(0.004)		
Citation (5)	0.014***			0.005*	0.013***	0.870	7548
	(0.004)			(0.003)	(0.005)		
Citation (6)		−0.055***		0.005	0.015***	0.874	7434
		(0.016)		(0.003)	(0.005)		
Citation (7)	0.030***	−0.063***		0.005*	0.015***	0.874	7354
	(0.006)	(0.017)		(0.003)	(0.005)		
Citation (8)			−0.015**	0.004	0.015***	0.873	7354
			(0.007)	(0.003)	(0.005)		
Originality (9)	0.011***			0.005*	0.014***	0.872	7548
	(0.004)			(0.003)	(0.005)		
Originality (10)		−0.021*		0.004	0.015***	0.873	7434
		(0.012)		(0.003)	(0.005)		
Originality (11)	0.018***	−0.026**		0.005	0.016***	0.873	7354
	(0.005)	(0.013)		(0.003)	(0.006)		
Originality (12)			−0.003	0.004	0.016***	0.873	7354
			(0.005)	(0.003)	(0.006)		
Generality (13)	0.010**			0.004	0.016***	0.882	6814
	(0.005)			(0.003)	(0.006)		
Generality (14)		−0.059***		0.003	0.018***	0.885	6700
		(0.018)		(0.003)	(0.006)		
Generality (15)	0.028***	−0.067***		0.004	0.019***	0.885	6620
	(0.007)	(0.020)		(0.003)	(0.006)		
Generality (16)			−0.018**	0.003	0.019***	0.884	6620
			(0.008)	(0.003)	(0.006)		
R&D (17)	1.848**			3.833***	3.062***	0.695	3582
	(0.862)			(1.052)	(1.210)		
R&D (18)		−0.661		3.759***	3.387***	0.695	3561
		(2.157)		(1.045)	(1.282)		
R&D (19)	2.166**	−1.279		3.892***	3.361***	0.695	3536
	(0.908)	(2.243)		(1.077)	(1.291)		
R&D (20)			0.503	3.863***	3.359***	0.695	3536
			(1.030)	(1.077)	(1.292)		

5.3 Table 3: high-tech intensiveness and innovation

Read: Table 3 is the core test of Hypothesis 2.

- **Patent counts, row (3):** $\beta_1 = 0.038$, $\beta_2 = -0.096$. For the average high-tech industry, patent counts rise by about **3.0%** with greater equity development and fall by about **6.6%** with greater credit development.
- **Citations, row (7):** $\beta_1 = 0.030$, $\beta_2 = -0.063$. This gives about **2.4%** and **-4.3%** effects, respectively.
- **Originality, row (11):** $\beta_1 = 0.018$, $\beta_2 = -0.026$.
- **Generality, row (15):** $\beta_1 = 0.028$, $\beta_2 = -0.067$. This is one of the strongest quality/breadth results in the paper.
- **R&D, rows (17)–(20):** the equity coefficient is positive and significant in several specifications, while the credit coefficient is negative but insignificant.
- **Overall finance.** In contrast to Table 2, the overall-finance interaction is often *negative and significant* here, which the authors attribute to the strong negative role of credit markets in high-tech industries.

Why it matters. Table 3 shows that the paper's second mechanism is not redundant. The findings are not just about industries needing money; they are also about industries needing *the right kind* of money.

5.4 Table 4: country-industry fixed effects

Read: Table 4 asks whether the baseline results survive when the paper controls for persistent industry-specific heterogeneity inside each country.

- In Panel A (external finance dependence), the equity coefficients remain positive and significant and the credit coefficients remain negative and significant for all patent-based measures.
- In Panel B (high-tech), the same pattern holds for patent, citation, originality, and generality.
- Example: in Panel A, patent $\beta_1 = 0.006$ and $\beta_2 = -0.004$, both highly significant; in Panel B, patent $\beta_1 = 0.005$ and $\beta_2 = -0.004$, again highly significant.

Why it matters. This is the strongest "it is not some country-specific industry quirk" check.

5.5 Table 5: standard errors clustered only by country

Read: Table 5 redoes the main regressions clustering standard errors only at the country level.

- For external finance dependence, the main patent-based coefficients stay positive for equity and negative for credit.
- For high-tech intensity, the main patent-based coefficients also stay positive for equity and negative for credit.

Why it matters. The inference is not an artifact of the two-way clustering choice.

Table 4
Robustness checks – country-industry fixed effects.

This table reports the robustness check results with country–industry fixed effects controlled. We estimate $Innovation_{ijt+1} = \beta_0 + \beta_1$ ($Equity_{it} \times Industry_j$) $+ \beta_2$ ($Credit_{it} \times Industry_j$) $+ \beta_3$ $Value\text{-}Added_{ijt} + \beta_4$ $US\text{-}Export_{ijt} + \eta_{it} + \eta_{jt} + \epsilon_{ijt+1}$. $Innovation_{ijt+1}$ is the relative innovation measures in industry j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{it}$ denotes country i's stock market capitalization over GDP in year t, and $Credit_{it}$ denotes country i's bank credit over GDP in year t. $Industry_j$ equals *Dependence*$_j$ in Panel A and *High-tech*$_j$ in Panel B. *Dependence*$_j$ denotes industry j's dependence on external financing, and *High-tech*$_j$ denotes industry j's median R&D growth in the US. $Value\text{-}Added_{ijt}$ denotes the share of industry j's value-added in country i in year t. $US\text{-}Export_{ijt}$ denotes the share of industry j's in country i's export to the US in year t. η_{jt} denotes the dummies for industry j in country i. ϵ_{ijt+1} denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country and industry. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976–2006.

Panel A Innovation	$Equity_{it} \times Dependence_j$	$Credit_{it} \times Dependence_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.006***	−0.004***	−0.003	−0.014	0.892	7354
	(0.001)	(0.001)	(0.009)	(0.016)		
Citation	0.005***	−0.011***	0.002	−0.021	0.653	7354
	(0.001)	(0.002)	(0.010)	(0.023)		
Originality	0.004***	−0.006***	−0.007	−0.025	0.563	7354
	(0.001)	(0.002)	(0.012)	(0.027)		
Generality	0.007***	−0.006***	0.002	−0.028	0.744	6620
	(0.001)	(0.002)	(0.011)	(0.029)		
R&D	0.063**	−0.100*	10.254	−1.579	0.637	3536
	(0.027)	(0.053)	(14.360)	(61.731)		
Panel B Innovation	$Equity_{it} \times High\text{-}tech_j$	$Credit_{it} \times High\text{-}tech_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.005***	−0.004***	−0.005	−0.005	0.886	7354
	(0.001)	(0.001)	(0.007)	(0.014)		
Citation	0.003***	−0.010***	−0.013	−0.012	0.639	7354
	(0.001)	(0.002)	(0.009)	(0.024)		
Originality	0.004***	−0.010***	−0.013	−0.015	0.621	7354
	(0.001)	(0.002)	(0.009)	(0.024)		
Generality	0.002***	−0.012***	0.000	−0.014	0.667	6620
	(0.001)	(0.003)	(0.007)	(0.022)		
R&D	0.143	0.885	0.313	−4.324	0.886	3536
	(0.076)	(0.541)	(0.824)	(5.781)		

Table 5
Robustness checks – clustering standard errors by country.

This table reports the robustness check results with standard errors of coefficient estimates clustered by country. We estimate $Innovation_{ijt+1} = \beta_0 + \beta_1$ ($Equity_{it} \times Industry_j$) $+ \beta_2$ ($Credit_{it} \times Industry_j$) $+ \beta_3$ $Value\text{-}Added_{ijt} + \beta_4$ $US\text{-}Export_{ijt} + \eta_{it+1} + \mu_j + \epsilon_{ijt+1}$. $Innovation_{ijt+1}$ is the relative innovation measures in industry j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{it}$ denotes country i's stock market capitalization over GDP in year t, and $Credit_{it}$ denotes country i's bank credit over GDP in year t. $Industry_j$ equals *Dependence*$_j$ in Panel A and *High-tech*$_j$ in Panel B. *Dependence*$_j$ denotes industry j's dependence on external financing, and *High-tech*$_j$ denotes industry j's median R&D growth in the US. $Value\text{-}Added_{ijt}$ denotes the share of industry j's value-added in country i in year t. $US\text{-}Export_{ijt}$ denotes the share of industry j's in country i's export to the US in year t. η_{it+1} denotes the dummies for country i in year $t+1$, and μ_j denotes the dummies for industry j. ϵ_{ijt+1} denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country. ***, **, and * Denote statistical significance at the 1%, 5%, and 10% levels, respectively. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976–2006.

Panel A Innovation	$Equity_{it} \times Dependence_j$	$Credit_{it} \times Dependence_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.047**	−0.128***	0.006	0.055	0.836	7354
	(0.022)	(0.045)	(0.011)	(0.043)		
Citation	0.033**	−0.087**	0.011	0.028	0.767	7354
	(0.015)	(0.039)	(0.010)	(0.018)		
Originality	0.026**	−0.059**	0.012	0.033	0.775	7354
	(0.012)	(0.028)	(0.011)	(0.021)		
Generality	0.025**	−0.073***	0.010	0.039	0.804	6620
	(0.011)	(0.023)	(0.012)	(0.025)		
R&D	0.943*	−1.008*	45.516	20.709	0.239	3536
	(0.514)	(0.498)	(45.993)	(52.176)		
Panel B Innovation	$Equity_{it} \times High\text{-}tech_j$	$Credit_{it} \times High\text{-}tech_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.038**	−0.096***	0.003	0.025**	0.920	7354
	(0.018)	(0.024)	(0.006)	(0.012)		
Citation	0.030**	−0.063***	0.005	0.015**	0.874	7354
	(0.011)	(0.020)	(0.005)	(0.007)		
Originality	0.018***	−0.026**	0.005	0.016**	0.873	7354
	(0.006)	(0.010)	(0.004)	(0.006)		
Generality	0.028**	−0.067***	0.004	0.019**	0.885	6620
	(0.011)	(0.017)	(0.005)	(0.007)		
R&D	2.166	−1.279	3.892	3.361	0.695	3536
	(1.416)	(1.394)	(2.752)	(3.685)		

5.6 Table 6: alternative finance proxies

Read: Table 6 replaces the finance measures with:

- stock market *traded value*/GDP for equity-market development,
- *private credit*/GDP for credit-market development.

- Panel A, patent: 0.057 for equity and −0.132 for credit.
- Panel B, patent: 0.033 for equity and −0.094 for credit.

Why it matters. The results are not tied to one narrow measurement choice for financial development.

Table 6
Robustness checks – alternative proxies for financial development.

This table reports the robustness check results using alternative proxies for financial development. We estimate $Innovation_{ijt+1} = \beta_0 + \beta_1$ ($Equity_{it} \times Industry_j$) $+ \beta_2$ ($Credit_{it} \times Industry_j$) $+ \beta_3$ $Value\text{-}Added_{ijt} + \beta_4$ $US\text{-}Export_{ijt} + \eta_{it+1} + \mu_j + \epsilon_{ijt+1}$. $Innovation_{ijt+1}$ is the relative innovation measures in industry j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{it}$ denotes country i's stock market traded value over GDP in year t, and $Credit_{it}$ denotes country i's private credit over GDP in year t. $Industry_j$ equals *Dependence*$_j$ in Panel A and *High-tech*$_j$ in Panel B. *Dependence*$_j$ denotes industry j's dependence on external financing, and *High-tech*$_j$ denotes industry j's median R&D growth in the US. $Value\text{-}Added_{ijt}$ denotes the share of industry j's value-added in country i in year t. $US\text{-}Export_{ijt}$ denotes the share of industry j's in country i's export to the US in year t. η_{it+1} denotes the dummies for country i in year $t+1$, and μ_j denotes the dummies for industry j. ϵ_{ijt+1} denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country and industry. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976–2006.

Panel A Innovation	$Equity_{it} \times Dependence_j$	$Credit_{it} \times Dependence_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.057***	−0.132***	0.005	0.055***	0.834	7354
	(0.015)	(0.027)	(0.007)	(0.012)		
Citation	0.026**	−0.074***	0.009*	0.028**	0.765	7354
	(0.013)	(0.028)	(0.005)	(0.011)		
Originality	0.014	−0.045*	0.011*	0.033***	0.774	7354
	(0.010)	(0.023)	(0.006)	(0.012)		
Generality	0.008	−0.057**	0.009	0.040***	0.803	6620
	(0.012)	(0.029)	(0.007)	(0.013)		
R&D	0.000	−1.004	47.535	16.295	0.237	3536
	(0.843)	(1.565)	(43.459)	(37.142)		
Panel B Innovation	$Equity_{it} \times High\text{-}tech_j$	$Credit_{it} \times High\text{-}tech_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.033***	−0.094***	0.002	0.025***	0.920	7354
	(0.010)	(0.020)	(0.003)	(0.004)		
Citation	0.024**	−0.058***	0.004	0.016***	0.873	7354
	(0.010)	(0.020)	(0.003)	(0.006)		
Originality	0.003	−0.014	0.004	0.016***	0.873	7354
	(0.007)	(0.014)	(0.003)	(0.006)		
Generality	0.017*	−0.062***	0.003	0.019***	0.885	6620
	(0.010)	(0.019)	(0.003)	(0.006)		
R&D	−0.873	0.105	3.781***	3.482***	0.695	3536
	(3.002)	(2.612)	(1.064)	(1.315)		

Table 7
Financial development, an alternative proxy for high-tech intensiveness, and innovation.

This table reports the robustness check results using an alternative proxy for an industry's high-tech intensiveness. We estimate $Innovation_{ijt+1} = \beta_0 + \beta_1$ ($Equity_{it} \times High\text{-}tech_j$) $+ \beta_2$ ($Credit_{it} \times High\text{-}tech_j$) $+ \beta_3$ $Value\text{-}Added_{ijt} + \beta_4$ $US\text{-}Export_{ijt} + \eta_{it+1} + \mu_j + \epsilon_{ijt+1}$. $Innovation_{ijt+1}$ is the relative innovation measure in industry j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{it}$ denotes country i's stock market capitalization over GDP in year t, and $Credit_{it}$ denotes country i's bank credit over GDP in year t. $High\text{-}tech_j$ denotes two-digit SIC industry j's median coefficient from regressing logarithmic ratio of market value to total assets on five-year R&D expenses scaled by total assets. $Value\text{-}Added_{ijt}$ denotes the share of industry j's value-added in country i in year t. $US\text{-}Export_{ijt}$ denotes the share of industry j's in country i's export to the US in year t. η_{it+1} denotes the dummies for country i in year $t+1$, and μ_j denotes the dummies for industry j. ϵ_{ijt+1} denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country and industry. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Our sample includes industries with two-digit SIC codes between 20 and 39. The sample period is 1976–2006.

Innovation	$Equity_{it} \times High\text{-}tech_j$	$Credit_{it} \times High\text{-}tech_j$	*Value-Added*	*US-Export*	*Adj-*R^2	*Obs.*
Patent	0.011**	−0.017*	0.007	0.062**	0.838	7354
	(0.005)	(0.009)	(0.013)	(0.024)		
Citation	0.003**	−0.005***	−0.004	0.026	0.774	7354
	(0.002)	(0.002)	(0.007)	(0.023)		
Originality	0.006*	−0.008*	0.002	0.029**	0.788	7354
	(0.003)	(0.004)	(0.008)	(0.014)		
Generality	0.002**	−0.003***	−0.003	0.039	0.817	6620
	(0.001)	(0.001)	(0.008)	(0.033)		
R&D	−0.043	0.013	−0.430	−0.265	0.154	3536
	(0.110)	(0.149)	(0.415)	(0.354)		

5.7 Table 7: alternative proxy for high-tech intensiveness

Read: Table 7 rebuilds the high-tech measure using how strongly financial markets value R&D at the industry level.

- Patent: $\beta_1 = 0.011$, $\beta_2 = -0.017$.

Table 8
Robustness checks – technology class analysis.

This table reports the robustness check results using innovation proxies defined at the technology class level. We estimate $Innovation_{i,j,t+1} = \beta_0 + \beta_1 (Equity_{i,t} \times Industry_j) + \beta_2 (Credit_{i,t} \times Industry_j) + \beta_3\ Value\text{-}Added_{i,t} + \beta_4\ US\text{-}Export_{i,t+1} + \eta_{i,t+1} + \mu_j + \epsilon_{j,t+1}$. $Innovation_{j,t+1}$ is the relative innovation measure in technology class j in country i in year $t+1$. Relative innovation measures are the count, citation, originality, or generality of non-government patents in year $t+1$ and R&D in year t scaled by the corresponding values in US data. $Equity_{i,t}$ denotes country i's stock market capitalization over GDP in year t, and $Credit_{i,t}$ denotes country i's bank credit over GDP in year t. $Industry_j$ equals $Dependence_j$ in Panel A and $High\text{-}tech_j$ in Panel B. $Dependence_j$ denotes technology class j's dependence on external financing, and $High\text{-}tech_j$ denotes technology class j's median R&D growth in the US. $Value\text{-}Added_{i,t}$ denotes the share of technology class j's value-added in country i in year t. $US\text{-}Export_{i,t}$ denotes the share of technology class j's in country i's export to the US in year t. $\eta_{i,t+1}$ denotes the dummies for country i in year $t+1$, and μ_j denotes the dummies for technology class j. $\epsilon_{j,t+1}$ denotes the error term. All regressions include a constant term (unreported). The statistical inferences are based on standard errors (reported in parentheses) clustered by country and industry. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The sample period is 1976–2006.

Panel A Innovation	$Equity_{i,t} \times Dependence_j$	$Credit_{i,t} \times Dependence_j$	*Value-Added*	*US-Export*	*Adj-R^2*	*Obs.*
Patent	0.059***	−0.089***	−0.446	1.525	0.166	63620
	(0.018)	(0.030)	(0.367)	(1.121)		
Citation	0.030***	−0.037***	−0.382	1.272	0.141	62138
	(0.011)	(0.013)	(0.315)	(1.021)		
Originality	0.042**	−0.044***	−0.393	1.467	0.167	52628
	(0.016)	(0.011)	(0.339)	(1.075)		
Generality	0.005	−0.033*	−0.565	1.848	0.159	34812
	(0.021)	(0.019)	(0.421)	(1.294)		
R&D	0.005	0.003	−1.979	5.696	0.414	62497
	(0.004)	(0.007)	(1.366)	(3.807)		
Panel B Innovation	$Equity_{i,t} \times High\text{-}tech_j$	$Credit_{i,t} \times High\text{-}tech_j$	*Value-Added*	*US-Export*	*Adj-R^2*	*Obs.*
Patent	0.622**	−1.988***	−0.387	1.287	0.167	63620
	(0.294)	(0.474)	(0.308)	(0.906)		
Citation	0.796**	−2.386***	−0.291	0.970	0.142	62138
	(0.395)	(0.691)	(0.231)	(0.730)		
Originality	0.477*	−1.679***	−0.339	1.228	0.168	52628
	(0.255)	(0.456)	(0.272)	(0.841)		
Generality	0.389	−1.528***	−0.481	1.519	0.160	34812
	(0.270)	(0.381)	(0.335)	(0.988)		
R&D	36.570*	−91.687**	−37.772	142.206	0.414	62497
	(18.686)	(42.162)	(36.234)	(106.300)		

- Citation: $\beta_1 = 0.003$, $\beta_2 = -0.005$.
- Originality: $\beta_1 = 0.006$, $\beta_2 = -0.008$.
- Generality: $\beta_1 = 0.002$, $\beta_2 = -0.003$.

Why it matters. The estimates are smaller and sometimes weaker statistically, but the direction is the same. So the results are not just an artifact of measuring high-tech status with R&D growth.

5.8 Table 8: technology-class analysis

Read: Table 8 redefines innovation at the three-digit technology-class level instead of the two-digit SIC industry level, expanding the sample to **428 technology classes**.

- Panel A (external finance dependence): patent 0.059 for equity and −0.089 for credit; citation 0.030 and −0.037.
- Panel B (high-tech): patent 0.622 for equity and −1.988 for credit; citation 0.796 and −2.386.

Why it matters. The core qualitative pattern survives when innovation is measured in a more patent-native classification system.

6 Short summary

- The paper asks whether financial development promotes innovation, but insists on separating **equity** from **credit**.
- It measures innovation with U.S. patent data and public-firm R&D, and uses U.S.-based industry characteristics to build a Rajan–Zingales type interaction design.
- Industries that need external finance more, and industries that are more high-tech intensive, innovate **more** in countries with stronger equity markets.
- The same industries innovate **less** in countries with deeper banking credit markets.

- The findings hold for patent quantity, citations, originality, and generality, and survive demanding robustness checks.
- The paper's central lesson is that the composition of financial development matters: innovation responds favorably to equity-market development and unfavorably, on average, to banking-credit development in precisely those sectors where innovation is hardest to finance.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that financial development does not have a single monotone effect on innovation. Instead, **equity-market development and credit-market development have systematically different effects**. Industries that are more externally finance dependent and more high-tech intensive experience disproportionately more innovation when equity markets are more developed, but less innovation when credit markets are more developed.

7.2 Economic intuition

The intuition is simple but important.

- Innovation is risky, long-horizon, intangible, and often hard to collateralize.
- Equity finance is naturally better matched to these properties because it tolerates upside-skewed payoffs, does not impose fixed repayment obligations, and generates information through prices.
- Credit finance is naturally less well matched because banks prefer safer, collateral-backed projects and can be conservative or rent-extracting in the face of innovative uncertainty.

This is why "financial development" is too crude a concept unless one decomposes it into its equity and credit components.

7.3 Takeaway

The right way to read the paper is:

- not "finance helps innovation,"
- but rather "**equity-market development helps innovation where outside finance and technological risk matter most, while bank-credit development can impede it in those same places**."

That is the paper's contribution to both the finance-growth literature and the finance-and-innovation literature.